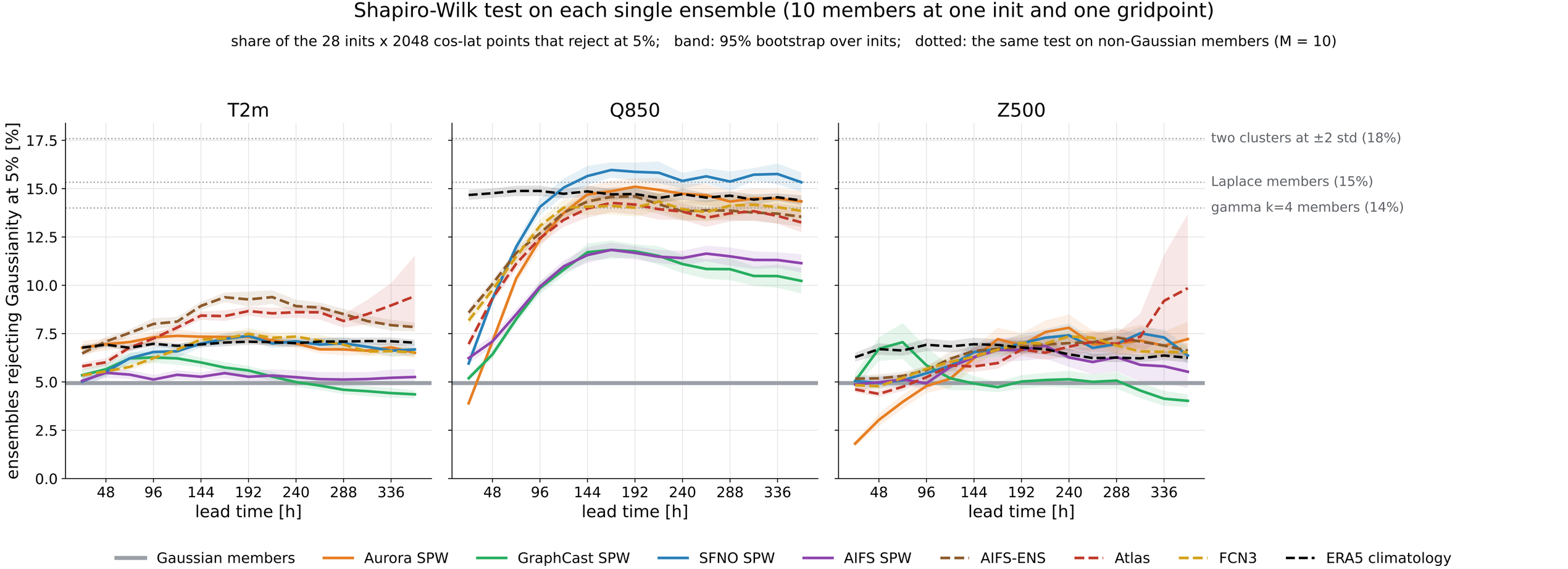
Shapiro-Wilk test on each single ensemble (10 members at one init and one gridpoint)
share of the 28 inits x 2048 cos-lat points that reject at 5%; band: 95% bootstrap over inits; dotted: the same test on non-Gaussian members (M = 10)
T2m
Q850
Z500
ensembles rejecting Gaussianity at 5% [%]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
48
96
144
192
240
288
336
lead time [h]
two clusters at ±2 std (18%)
Laplace members (15%)
gamma k=4 members (14%)
Gaussian members
Aurora SPW
GraphCast SPW
SFNO SPW
AIFS SPW
AIFS-ENS
Atlas
FCN3
ERA5 climatology